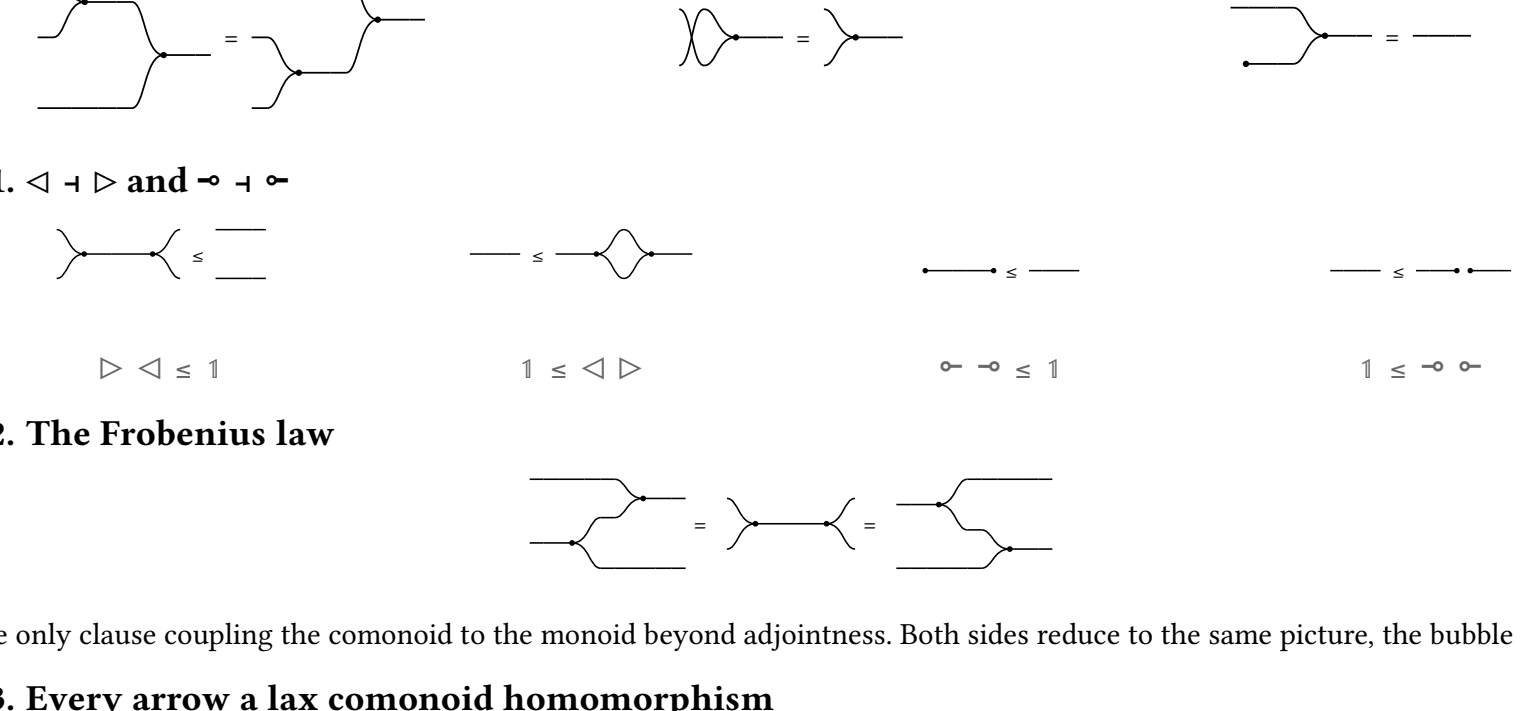


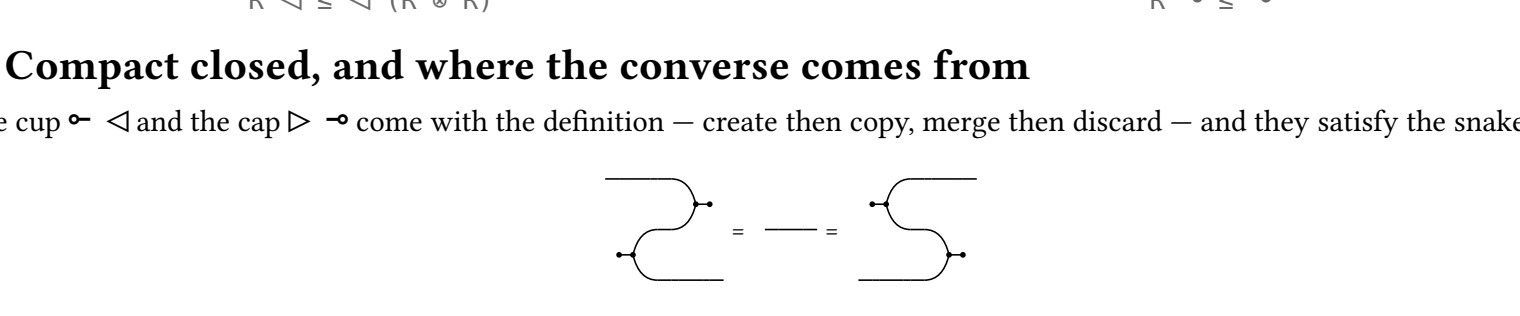
The allegory axioms, and what defines them in the Frobenius calculus

1. Cartesian bicategory of relations

$\langle \lhd, \rhd \rangle, \multimap, \dashv, \triangleright, \triangleleft, \circ, \circ^*$, commutative, **Frobenius**, arrows **lax**.



1.1. $\lhd \triangleleft \dashv \triangleright$ and $\rhd \triangleright \multimap \dashv \triangleleft$

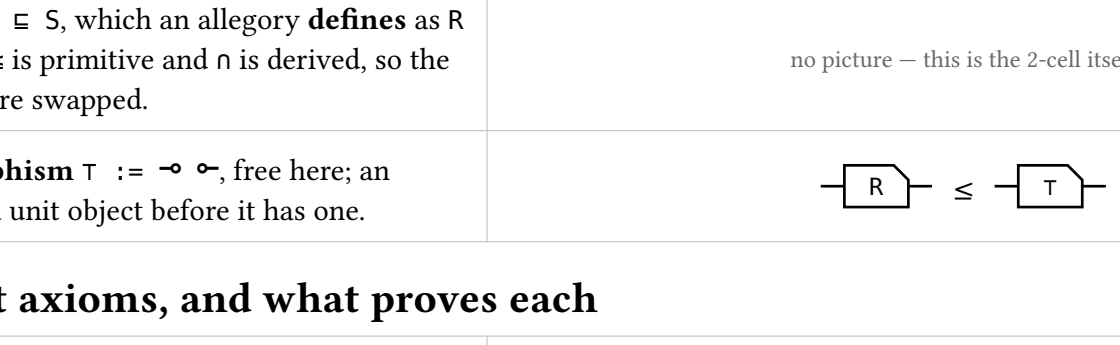


1.2. The Frobenius law



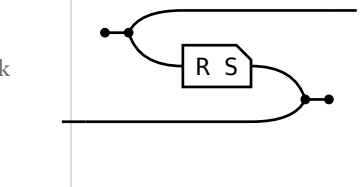
The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, the bubble $\triangleright \triangleleft$.

1.3. Every arrow a lax comonoid homomorphism



2. Compact closed, and where the converse comes from

The cap $\multimap \triangleleft$ and the cap $\triangleright \multimap$ come with the definition – create then copy, merge then discard – and they satisfy the snakes:



So every object is its own dual, and R^* is R with its wires bent round the cup and the cap. That is what makes reciprocity **definable** rather than primitive, and its four laws theorems:

- (i) $1^* = 1$
- (ii) $(R \ S)^* = S^* \ R^*$
- (iii) $(R \otimes S)^* = R^* \otimes S^*$
- (iv) $R \leq S$ implies $R^* \leq S^*$

(i) is the snake above; (ii) is drawn below. The other two say nothing a picture makes clearer.

3. The allegory primitives are definitions here

primitive, and what defines it here	picture
reciprocity $R^* := \triangleleft (R \otimes 1) \text{ cap}$, not a generator. It is the cup and the cap that reverse the arrow – a wire merely hopped over the box and back down is an isotopy of R itself.	
intersection $R \cap S := \triangleleft (R \otimes S) \triangleright$. Copy the inputs, run both, merge; the merge forces the two results to agree.	
containment $R \sqsubseteq S$, which an allegory defines as $R \cap S = R$. Here $\sqsubseteq$ is primitive and $\cap$ is derived, so the two directions are swapped.	no picture – this is the 2-cell itself
maximal morphism $\top := \multimap \circ^*$, free here; an allegory needs a unit object before it has one.	

4. The eight axioms, and what proves each

axiom, and what supplies it	picture
$(R^*)^* = R$ – the snake. Bending down and back up straightens out, so bending twice is the identity.	
$(R \ S)^* = S^* \ R^*$ – mirroring the picture. Reflecting a chain left to right reverses it; each box comes back flipped.	
$(R \cap S)^* = R^* \cap S^*$ – the same mirroring. $\triangleleft$ and $\triangleright$ are each other's mirror image, so reflecting the convolution leaves its shape alone and only turns the boxes round.	
$R \cap R = R$ – the lax copy law and the special law. Copy, run R twice, merge – the lax copy law pushes R back through the copy and $\triangleleft \triangleright = 1$ closes the bubble.	
$R \cap S = S \cap R$ – cocommutativity and commutativity. The two strands of $\triangleleft$ are interchangeable, and so are those of $\triangleright$.	
$R \cap (S \cap T) = (R \cap S) \cap T$ – coassociativity and associativity. Either bracketing builds the same three-way copy tree, and dually the same merge tree.	
$R \cap (S \cap T) \sqsubseteq R \cap S \cap T$ – the lax copy law. Running R and then copying is below copying and then running R on each strand.	
$R \cap S \cap T \sqsubseteq (R \cap S) \cap T$, the modular law, has no adjoined as an axiom by an allegory, which has no structure to derive it from. Here Frobenius and the lax copy law give it. Strip the shared prefix $\triangleleft (R \otimes T)$ off both sides and what is left is the merge slide: a box slides through a merge at the cost of its converse on the other strand.	

5. Unitary and pre-tabular – what the other direction needs

The eight axioms are all a cartesian bicategory has to give back. Going the other way needs two further conditions.

condition	allegory	Frobenius side
unitary	T is a unit when $1 \ T$ is the largest endomorphism on it and every object carries an entire arrow to it. The point of a unit is that it makes $T \circ \text{unit} \Rightarrow \text{id}$, $\text{id} \Rightarrow T \circ \text{unit}$ through the unit – which is also where the lax discard law comes from.	already there, for free. The monoidal unit 1 is the unit object, $\multimap : n \rightarrow 1$ is the entire arrow every object carries to it – $1 \leq \multimap$ says so – and $T = \multimap \circ^*$ is maximal. Every cartesian bicategory of relations is unitary, with nothing assumed.
pre-tabular	f, g tabulate R when both are maps, $R = f^* \circ g$, and the two agree only on the diagonal: 	this is what $\otimes$ is. Given unitarity the condition reduces to tabulating each T , and a tabulation of T is precisely an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

Together they are the whole gap: cartesian bicategory of relations $\approx$ unitary pre-tabular allegory.

6. Union and residual

$\cup$ and $/$ are past where the eight axioms stop and past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure, a residual a **second composition** with linear adjoints. Both are built, so both draw.

A union is a **tape**: the rounded wrapper is the second monoidal product, and its $\triangleright \triangleleft$ open and close a branch a particle takes exactly one of – the same fork-runs-join shape as a meet, on a different product.

A residual is drawn as **long division**, deliberately not as its own definition. As a term it is a composition against a complement, and drawing it that way needs a complement to mean anything. The operation does not: it is one Galois connection, and the last section draws it with none.

statement	picture
$R \sqsubseteq R \cup S$, union $R \ S := (R \ S) [1, 1]$ offer both branches, then forget which was taken.	
$R \cup S = S \cup R$. The two branches of a tape are interchangeable, exactly as the two strands of $\triangleleft$ are.	
$1 \cup R = R$. 1 is created through the zero object; there is no shape for it, so it prints as a box.	
$R \ (S \cup T) = R \ S \cup R \ T$ – composition distributes over union, which $\cap$ does not. This is what a distributive allegory has over an allegory.	
$(R \cup S)^* = R^* \cup S^*$. The converse is a 2-functor for the tape layer too.	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$, the law that makes the whole distributive. The one picture with both layers in it: meets inside a tape on the left, tapes around meets on the right.	
$(R \ S) \ S \sqsubseteq R$ – the residual is a lower bound of the arrows it is the greatest of. The bar in the last section's amber ground for the numerator, the divisor laid inside it, a hairline of slack past it.	

7. $(R \ S)^* = S^* \ R^*$, drawn

The pictures below are not a retelling: each is drawn from one term of the chain that was actually checked, so a picture and the term beside it cannot drift apart.

7.1. These are not pictures of a converse

The converse is drawn with two bends, a cup on the way in and a cap on the way out. The proof never draws that. An arrow is the converse of R as soon as it satisfies one equation between UNBENT arrows into 1 – two strands in on the left, one cap on the right – and bending is a bijection, so settling the unbent equation settles the bent one. Hence every row below has a cap and no cup, and an arrow that "is a converse" shows up not as a bent box but as a box sitting on the **bottom** strand, mirrored.

the slide – the one rule the proof uses, and it uses it twice
A box on the BOTTOM strand facing a cap is the same as that box on the TOP strand, upright. Sliding round the bend is what turns R^* into R , and it is the entire content of the theorem – the other seven steps are bookkeeping.

7.2. The chain

One row per term: the right-hand side of every step is the left-hand side of the next. Watch which **strand** each box is on – bottom means mirrored, top means upright.

term	picture	the rule that reaches it
$(R \ S)^* = S^* \ R^*$		
$(1 \otimes (S^* \ R^*)) \text{ cap}$		the start. What has to be shown about S^* R^* : both boxes on the bottom strand, mirrored.
$= ((1 \otimes S^*) ((1 \otimes R^*) \text{ cap}))$		1. $\otimes$ is a functor. Same picture.
$= ((1 \otimes S^*) ((R \otimes 1) \text{ cap}))$		2. the slide. R^* is the box nearest the cap: it leaves the bottom strand, arrives on the top, and comes back upright as R . The first of the two steps that prove anything.
$= ((1 \otimes S^*) (R \otimes 1)) \text{ cap}$		3. re-bracketing. Same picture.
$= (R \otimes S^*) \text{ cap}$		4. interchange. R is on the top strand and S^* on the bottom, so running them in sequence and turning them side by side are the same thing. Shows as a re-spacing only.
$= ((R \otimes 1) (1 \otimes S^*)) \text{ cap}$		5. interchange back. Split them apart again. R first, so S^* is the box facing the cap.
$= (R \otimes 1) ((1 \otimes S^*) \text{ cap})$		6. re-bracketing. Same picture.
$= (R \otimes 1) ((S \otimes 1) \text{ cap})$		7. the slide. on S this time. Both boxes are now on top, both upright, in the order R then S . The second and last real step.
$= ((R \otimes 1) (S \otimes 1)) \text{ cap}$		8. re-bracketing. Same picture.
$= ((R \ S) \otimes 1) \text{ cap}$		9. interchange once more; the two top-strand boxes merge into one. This is R unbent. Done.

What the chain measures. Steps 1, 3, 6, 8 and 9 change nothing and step 4 changes only spacing, while the term column changes at every row. Those are associative, the unit laws and interchange – precisely the laws a string diagram has built into its geometry rather than as rewires. Seven of the nine steps are bureaucracy the picture language does not charge for, and the theorem is two slides.

8. The modular law, from Frobenius

Four terms, of which exactly one step is an inequality.

term	picture	the rule that reaches it
$(R \ S) \cap T$		the start: $R \ S \cap T$. A copy and a merge with $R \ S$ above and T below.
$= \triangleleft (R \otimes T) \triangleright ((S \otimes 1) \text{ cap})$		Factor out the shared prefix. $(R \ S) \otimes T$ is $(R \otimes T) (S \otimes 1)$, so everything before the S is common to both sides. An equality, and drawn as one.
$\leq \triangleleft (R \otimes T) \triangleright ((1 \otimes S^*) \triangleright S)$		The merge slide, under the shared prefix. S slides through the merge and reappears as S^* on the other strand. The only inequality in the proof.
$= (R \cap T) (S^*) \ S$		Fold the prefix back in. $R \otimes (T \ S^*)$ is $(R \otimes T) (1 \otimes S^*)$, and the copy-merge pair reads as a meet again. $(R \cap T) (S^*) \ S$, the goal.

8.1. Inside the middle step

The merge slide mentions only $S - R$ and T sit untouched in the prefix throughout. S occurs once on the left and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box, so it has to be the inequality step. The chain shows it is the only one.

term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.
$= ((S \otimes 1) \otimes 1) (1 \otimes \text{cap})$		Rebuild the merge from a copy and a cap. Which puts the term into the shape $(S \otimes 1) \otimes 1$ the lax copy law applies $S \triangleleft$.
$\leq ((S \otimes 1) \otimes 1) (1 \otimes \text{cap})$		The lax copy law: $S \triangleleft$ becomes $\triangleleft (S \otimes 1)$; the box is copied. The whole of the modular law, in one step.
$= (1 \otimes S^*) \triangleright S$		Reshape back, and the surviving duplicate slides round the cap to become S^* .

What the associators cost. α and ρ have no shape in this calculus and print as opaque boxes, so the two middle terms are less legible than the ends. That is the honest price of drawing what was checked; a hand-drawn version would have hidden them by working up to coherence.

9. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$. Applying (ii) and (iv) above gives the merge form and the unit form, the same statements about $\triangleright$ and $\circ^*$. Those four hold for every arrow. Their four REVERSES does not, and each one holding is a property of the arrow.

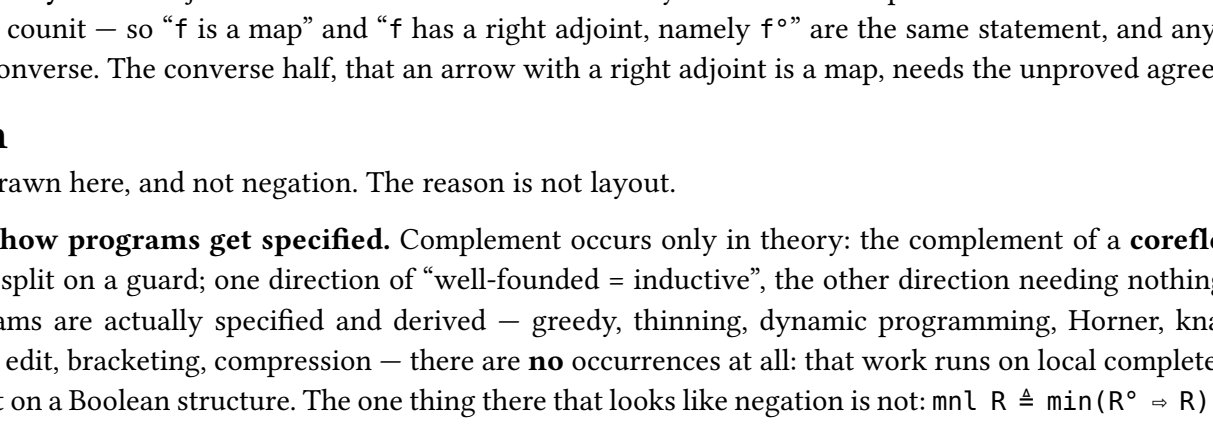
holds for every arrow	property	holds iff
$R \triangleleft S \leq \triangleleft (R \otimes R)$ 	(SV) single valued $R^* \ R \sqsubseteq 1$	
$R \multimap S \leq \multimap$ 	(TOT) total $1 \sqsubseteq R \ R^*$, With single valued, R is a map.	
$\triangleright \triangleleft S \leq (R \otimes R) \triangleright$ – the converse of the lax copy law. Not stated; see below.	(INJ) injective $R^* \ R^* \sqsubseteq 1$	
$\multimap \ R \leq \multimap$ – the converse of the lax discard law. Not stated; see below.	(SUR) surjective $1 \sqsubseteq R^* \ R$, that is R^* is entire.	

The right-hand column is the ADJOINT form of each property, and it is the definition used here, because it is literally the allegory's own *Simple and Entire* – so "f is a map" and "f has a right adjoint, namely f^* " are the same statement, and any total adjoint of a map must be its converse. The converse half, that an arrow with a right adjoint is a map, needs the unproved agreement above.

Why the merge form and the unit form have no picture. Both are converse-images, and getting them needs $\triangleleft^* = \triangleright$, $\multimap^* = \multimap$ and $(f \otimes g)^* = f^* \otimes g^*$. The first is a Frobenius computation at a COMPOSITE object, which is the clause the printed definition omits – the same gap that leaves the agreement above unproved. The four reverse inequations are the cause.

10. Entireness of an intersection

The domain of R is the part of the identity where R is defined. Dom $R = 1 \cap R \ R^*$; R is **entire** when it is defined everywhere, Dom $R = 1$, equivalently $1 \sqsubseteq R \ R^*$ – which is (TOT) above. When is an intersection entire?



Total $(R \cap S) \cap 1 \leq 1 \leq S^* \ R^*$

On the left $R \cap S$ is named **twice**; on the right R and S are named **once each** and the meet is gone. That is the point of the lemma – the meet is needed to ask whether something is entire, not to answer it – and it is why every later use quotes the right-hand form.

Where $\sqsubseteq$ is defined as $X \cap Y = X$, the two sides are the same proposition and there is nothing to prove. Here $\sqsubseteq$ is primitive, so both directions are real steps – and they cost very different things.

cancel the law is $(R/S) \ S \sqsubseteq R$ and not an equality – and it is a hairline on purpose, because R / S is the **largest** such quotient: any wider and S stops fitting.

Solid is what the equation pins: R 's left edge, S 's left edge, and the stretch before S , which is what R / S measures. Dashed is the slack: nothing to the right of S is determined by $T \ S \sqsubseteq R$.

Read it as a **measurement**, not as a containment of relations. $R : a \multimap c$ and $S : b \multimap c$ are not in the same hom-set, so S is not a sub-relation of R and neither is $R / S \multimap$ what sits inside R is the composite $(R/S) \ S$, which does have R 's type. Pushed too far, the metaphor breaks in the usual place: $\perp / \perp \neq T$.

the slack: why $\multimap$

transcribed – the universal property has a statement, this metaphor does not

the ten laws, their pictures, and – where the metaphor reaches – the same law as long division

Division is the right adjoint of composition: $T \sqsubseteq R/S \iff T \ S \sqsubseteq R$

$T \sqsubseteq S/R \iff S \ T \sqsubseteq R$

$S/R \ \& \ (R'/S')^{\perp}$: left division is right division conjugated by the converse, not a second primitive.

cancel: $(R/S) \ S \sqsubseteq R$ and $S \ (S/R) \sqsubseteq R$

The count of each adjunction: the quotient composed back with the divisor lands under the numerator, never on it – law the figure above draws.

associate: $R/(S : S_2) = (R/S_2) / S_1$ and $S \ T \sqcap R = T \ (S \sqcap R)$

Dividing by a composite one factor at a time. The composition is drawn on the left of each equation and swallowed into a label on the right – and the bar shows what that swallowing hides: two tiles laid where there was one.